RAJALAKSHMI ENGINEERING COLLEGE

Rama Raghavi T 250701582-2025-CSE

R2

Started on	Tuesday, 15 September 2026, 12:17 AM
State	Finished
Completed on	Tuesday, 15 September 2026, 12:17 AM
Time taken	42 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

// Function to find the upper bound (first element strictly greater than target)
int upper_bound(int *arr, int low, int high, int target) {
    while (low < high) {
        int mid = low + (high - low) / 2;
        if (arr[mid] <= target) {
            low = mid + 1;
        } else {
            high = mid;
        }
    }
    return low;
}

// Function to find the length of the Longest Non-decreasing Subsequence
int lengthOfLNDS(int *nums, int n) {
    if (n <= 0) return 0;

    // tails[i] stores the smallest tail of all non-decreasing subsequences of length
    int *tails = (int *)malloc(n * sizeof(int));
    int len = 0;

    for (int i = 0; i < n; i++) {
        int x = nums[i];

        // Find the first element strictly greater than x
        // Use upper_bound because duplicates (equal values) are allowed in non-decrea
        int idx = upper_bound(tails, 0, len, x);

        tails[idx] = x;

        // If x is greater than or equal to all elements in tails, len increases
        if (idx == len) {
            len++;
        }
    }

    free(tails);
    return len;
}

int main() {
    int n;

    // Read the number of elements
    if (scanf("%d", &n) != 1) return 0;

    int *sequence = (int *)malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
```

	Input	Expected	Got	
✔	9 -1 3 4 5 2 2 2 2 3	6	6	✔
✔	7 1 2 2 4 5 7 6	6	6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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